Rayan Aslam

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Education

Rutgers New Brunswick Honors College — New Brunswick, NJ
B.S. in Computer Science and Data Science — GPA: 3.8 — *Expected June 2028*
- Related Coursework: Data Structures & Algorithms, Computer Architecture, Linear Algebra, Discrete Math

Experience

Undergraduate Research Assistant — June 2026 – Present
Rutgers Art and AI Lab — *New Brunswick, NJ*
- Engineered **Python** automation pipelines using **NumPy** and **Pandas** to extract, aggregate, and structure visual statistics from 100+ artworks into analysis-ready datasets, accelerating downstream research workflows
- Applied **Hugging Face** models to extract deep feature representations across AI-generated and authentic artwork, enabling quantitative comparison of stylistic patterns and generative model behavior
- Developed reproducible analysis workflows for matrix construction and feature trend visualization using **Matplotlib**, producing publication-quality figures to support ongoing research findings

Web Developer — Jan 2026 – Present
Casual Harmony Acapella — *New Brunswick, NJ*
- Built and maintained the organization's official website using **React, Vite, and Tailwind CSS**
- Designed polished user interfaces with emphasis on usability and consistent visual branding across devices
- Developed functional contact systems to streamline communication between the organization and prospective members, clients, and audiences

Projects

Rubik's Cube Solver | *Python, PyTorch, Pandas, NumPy*
- Developed a 2×2 Rubik's Cube simulator in **Python** with full move set and scramble generation, producing **50k+ labeled states** to train machine learning models
- Designed and trained a **custom CNN in PyTorch** with residual blocks, dropout, and weight decay, achieving **80%** move-prediction accuracy and full-solve success rate on 5-move scrambles
- Built a **high-throughput data pipeline** using PyTorch Dataset/DataLoader with GPU preprocessing and batching, reducing data prep time and enabling rapid scaling of experiments
- Created evaluation tools for accuracy/loss tracking and benchmarking across repeated trials

Math Battles | *React, Node.js, Socket.io, Gemini API*
- Built a real-time multiplayer trivia game with timed rounds, live scoreboards, and persistent score tracking, synchronizing game state across all connected clients via bidirectional **Socket.io** event-driven messaging
- Integrated the **Gemini API** to dynamically generate calculus question-and-answer pairs at runtime, eliminating static question banks and delivering a unique, infinitely replayable experience each session
- Designed and shipped a complete full-stack application with a **React** frontend and **Node.js** backend end-to-end within a hackathon time constraint, demonstrating rapid prototyping and cross-functional execution under pressure

Leadership

Assistant Business Manager — April 2026 - Present
Casual Harmony Acapella — *New Brunswick, NJ*
- Organized concerts by coordinating venues, scheduling, logistics, marketing, and cross-team communication
- Directed financial operations by handling budgeting, reimbursements, and fundraising

Board Member — Jan 2026 - Present
Muslim Tech Collaborative — *New Brunswick, NJ*
- Organized a multi-university hackathon, coordinating logistics for 100+ participants and locating sponsors

Technical Skills

Languages: Java, Python, SQL, JavaScript, HTML/CSS, C/C++
Frameworks/Libraries: React, Node.js, PyTorch, pandas, NumPy, Matplotlib
Developer Tools: Git, GitHub, VS Code, PyCharm, IntelliJ